Safe Reinforcement Learning for Self-Adaptive Systems - Project Seminar – Final Report

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ABSTRACT

One significant challenge in traffic control systems is managing unprotected right turns at intersections, where the absence of dedicated turning signals can create collision risks. In this report we explore how to utilize reinforcement learning algorithms to dynamically adjust traffic light phases for mitigating collision risks while optimizing traffic flow. We use Simulation of Urban Mobility (SUMO) to simulate intersection traffic, integrating RL algorithms such as Deep Q-Learning (DQN), Advantage Actor Critic (A2C), Proximal Policy Optimization (PPO), and Trust Region Policy Optimization (TRPO) through SUMO-RL. In the report we investigate the impact of RL algorithms and speed limits on safety and efficiency, evaluating performance based on collision occurrences and time loss metrics. Results show TRPO's superior safety performance, while A2C exhibits the smallest time loss. Further analysis shows the influence of speed limits on collision types. Additionally, we introduce a shield algorithm to enhance safety in A2C agents, observing a trade-off between safety improvement and performance degradation. The report concludes with avenues for future research, including optimizing shield distances and incorporating safety into RL rewards.

1 INTRODUCTION

Motivation and Context. In current urban environments, traffic control plays a crucial role in regulating vehicle flow and ensuring road safety. Among the many challenges faced by traffic management systems, one prominent one is the management of unprotected right turns at intersections, where the absence of a dedicated turning signal can lead to potential collisions. Addressing this issue requires complex structures capable of dynamically adjusting traffic light phases to mitigate collision risks while optimizing traffic flow. In recent years, reinforcement learning (RL) has emerged as a promising approach for developing dynamically adaptive systems. RL algorithms may enable these systems to autonomously learn and adapt the decision making based on real-time traffic conditions and performance feedback. When optimized for safety, traffic control systems managed by RL agents can potentially reduce the number of accidents by regulating movement on intersections, while keeping traffic flow as high as possible.

Technology To simulate the intersection and traffic, we utilize the traffic simulation package Simulation of Urban Mobility (SUMO) [1]. The reinforcement learning algorithms used in our work include Deep Q-Learning (DQN) [2], Advantage Actor Critic (A2C) [3], Proximal Policy Optimization (PPO) [4] and Trust Region Policy Optimization (TRPO) [5]. Serving as an interface between SUMO and these RL algorithms we use SUMO-RL [6]. Our source code used in this report is made freely available[1].

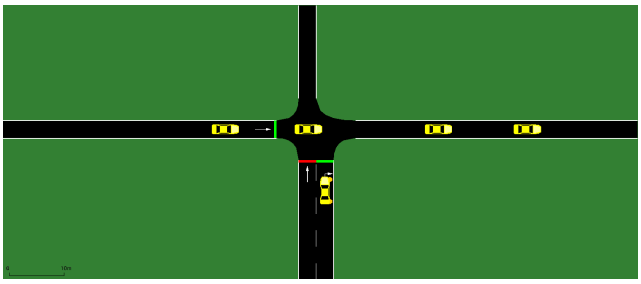


Figure 1: Sumo Snapshot

2 EXPERIMENTS

Road Network. The simulation of the traffic light at an intersection is setup within the Sumo simulator as seen in figure 1. The incoming and outgoing streets connected to the intersection are each 200 meters in length. In total, there are three lanes: the horizontal lane (H) enabling traffic flowing from west to east, the vertical control lane (Vc) accommodating traffic from south to north, and the vertical right turn lane (Vrt) serving traffic from south to east. Cars turning right on Vrt merge onto the same street as H, potentially leading to collisions starting at the intersection itself. Within each instance of the simulation a uniform speed limit is applied across all lanes. The speed limits were chosen according to common German regulations, including 30 km/h for residential areas, 50 km/h for urban traffic, 80 km/h for large vehicles such as trucks and buses, 100 km/h for intercity roads, and 130 km/h as the advisory speed limit on the autobahn.

Traffic Light Control. The unprotected right turns are regulated through specific phases in the traffic light system. The traffic light control agent chooses between three actions: permitting traffic solely on H, both vertical lanes, or enabling the unprotected right by giving a green light to H and Vrt concurrently. To be able to simulate collisions, vehicles making right turns ignore traffic on H entirely, entering the intersection immediately after receiving a green traffic light signal. The agent is setup to make decisions every 5th simulation step, directly adjusting the lights without yellow phases in between whenever an action is taken. In each experiment,

[1] github.com/hpi-sam/Safe-RL-4-SAS/tree/safe-rl-report

the traffic light control is managed by a reinforcement learning agent. For each combination of algorithm (A2C, DQN, PPO, TRPO) and speed limit (30, 50, 80, 100, 130), we trained the agent using Sumo-RL for 50 episodes, each consisting of 10000 time steps in the simulator environment. The traffic during training is probabilistically generated based on predefined traffic patterns, including balanced traffic on each lane and unbalanced traffic scenarios, with one lane having a higher volume of traffic compared to others, applied once to each lane. During training the agents are optimized for throughput, without taking safety into account.

Shield. The shield algorithm processes input parameters, including a specified distance, the status of the horizontal lane traffic, and the agent's intended action. If a vehicle is present on the horizontal lane within the shield distance from the intersection, the agent's action is checked. In cases where the action is deemed potentially unsafe, given by simultaneous green lights for both H and Vrt, it is replaced with a safe action where only H is given a green light, while Vrt is set to red. Shield distances tested include 0m (no shield), 1m (minimal shield), 5m, 10m, 20m, 50m and 100m. The agent is unaware of the shield and training is conducted with a shield distance set to 0m.

Evaluation. The evaluation is consistent across all experiments, using an identical traffic pattern. Vc experiences minimal traffic (1% probability of a car spawning per time step), while both H and Vrt exhibit balanced traffic (16% per lane). The simulation is run 300 times per experiment. Safety is assessed through occurrences of lateral and rear-end collisions. Furthermore, the quality of the agent is measured based on the "timeLoss" metric, representing the average time lost due to driving slower than desired [7].

3 RESULTS

Algorithms Figure 2 shows the comparison of safety metrics across the different reinforcement learning algorithms. According to the number of rear-end and lateral collisions, TRPO exhibits superior performance. A2C and DQN, when safety is not explicitly considered during training, show tendency towards riskier decisions. However, examining Figure 3 reveals that A2C excels in terms of the time loss value. TRPO and PPO show moderate performance, while DQN appears comparatively worse than the alternatives. A2C and TRPO are worth further investigation, as they excel either in performance or safety metrics, respectively. DQN and PPO are excluded from additional evaluation in our experiments.

Speed Limits Figures 4, 5 and 6 demonstrate the generalizability of these findings across various speed limits. A2C consistently outperforms TRPO in terms of the time loss value, irrespective of the trained and evaluated speed limit. A2C shows consistent results, indicated by the minimal variance observed in the boxplot. Conversely. TRPO displays erratic behavior at higher speeds. This effect can also be examined while evaluating the safety scores. As anticipated, TRPO demonstrates superior safety performance compared to A2C across most speed limits. However, while collisions are nearly absent at a speed limit of 100 km/h there is a sudden unexpected surge in collisions with further increases in speed limit. Notably, an interesting observation is the reduction of lateral collisions with higher speed limits, while the number of rear-end collisions remains close to unchanged for the A2C data.

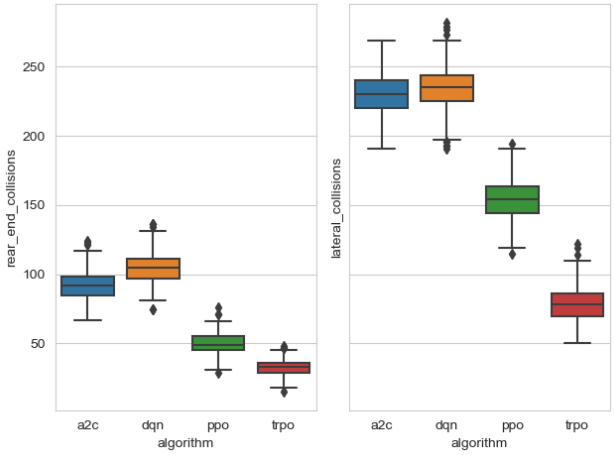


Figure 2: Collisions per Algorithm

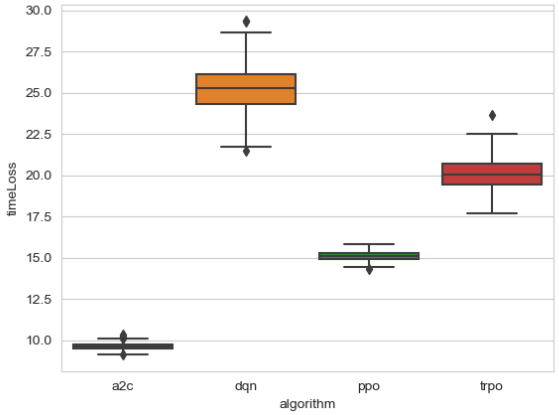


Figure 3: Time Loss per Algorithm

We think that this phenomenon arises from vehicles clearing the intersection faster, reducing the likelihood of collisions overall, particularly for lateral collisions. However, the increase in speed also extends braking distances, increasing the probability of rear-end collisions. These opposing effects appear to balance each other out.

Shield. Given A2C's strong performance in the context of time loss while having an apparent deficiency in safety measures, we decided to enhance the A2C agent with a shield. Figures 7, 8 and 9 illustrate the noticeable impact of shield distance on both agent safety and performance. With increasing shield distance, both lateral and rear-end collisions exhibit a consistent decline. The data does not indicate different effects depending on collision type. However, the shield adversely affects performance, evident by a rise in the time loss value and an increase in variance, showing the occurrence of exceptionally high time loss values. There exists an evident trade-off between safety and performance when taking

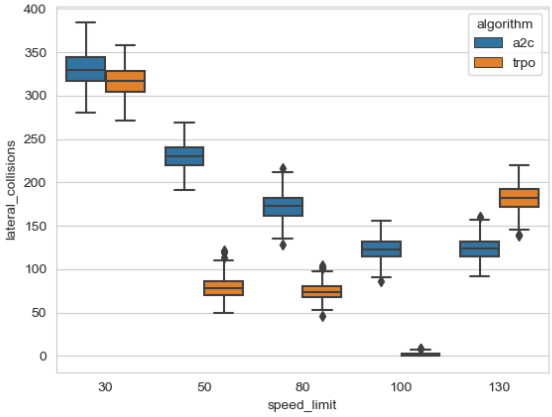


Figure 4: Lateral Collisions over Speed Limit

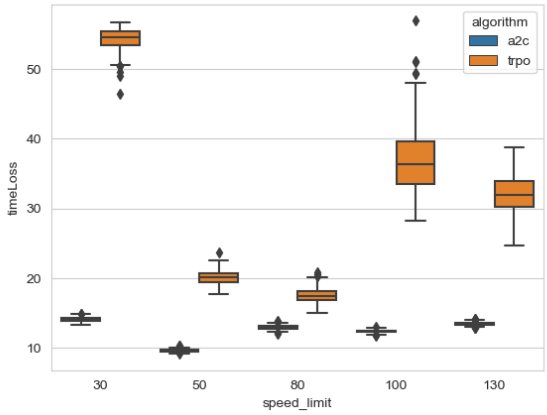


Figure 6: Time Loss over Speed Limit

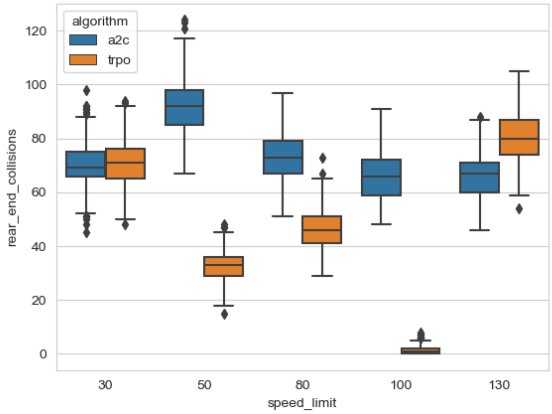


Figure 5: Rear End Collisions over Speed Limit

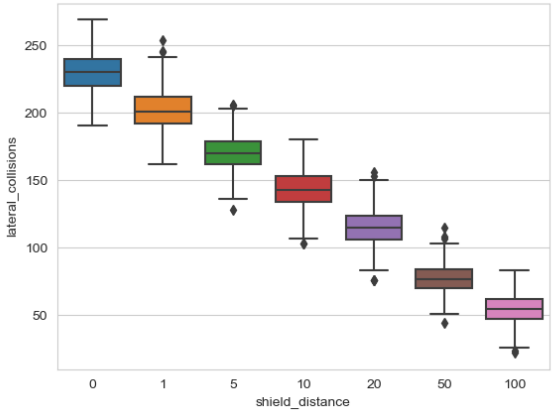


Figure 7: Lateral Collisions over Shield Distance

action through establishing a shield distance. Depending on the circumstances, an appropriate shield distance can be dynamically selected to align with current requirements.

4 DISCUSSION

Conclusion. Our experiments and evaluation have shed light on various aspects of traffic management within the Sumo environment, particularly focusing on the effectiveness of different algorithms in an unprotected right traffic light scenario. Our analysis has identified notable performance variations among the reinforcement learning algorithms, with A2C demonstrating recommendable performance metrics in terms of time loss but lacking in safety considerations. The introduction of a shield to augment the A2C agent revealed a trade-off between safety enhancement and performance degradation, highlighting the complexity of balancing these competing priorities. Additionally, our examination of collision types at different speed limits underscores the complexity of traffic dynamics, where higher speeds pose greater risks despite potentially mitigating certain collision types.

Future Work. Based on the findings presented in this report, several avenues are worth exploring further. Achieving an optimal balance between safety and performance could involve conducting a grid search to determine the ideal shield distance. Moreover, investigating the training and evaluation of agents across varying speed limits is interesting. For instance, training an agent on a specific speed limit and subsequently evaluating its performance and safety on slower or faster speed limits would provide valuable insights into the impact of speed variations on agent behavior. Directly incorporating safety consideration into the reward structure of the reinforcement learning agent is another promising direction, possibly through the implementation of penalty values deducted upon

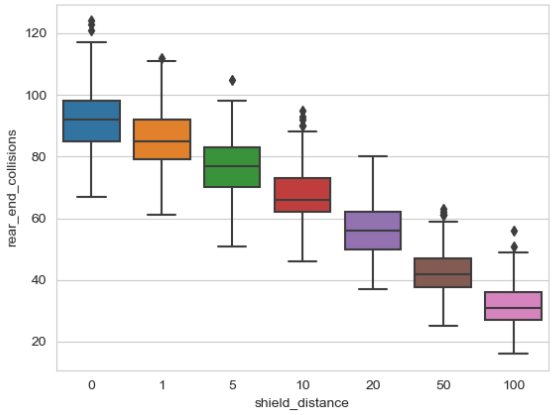


Figure 8: Rear End Collisions over Shield Distance

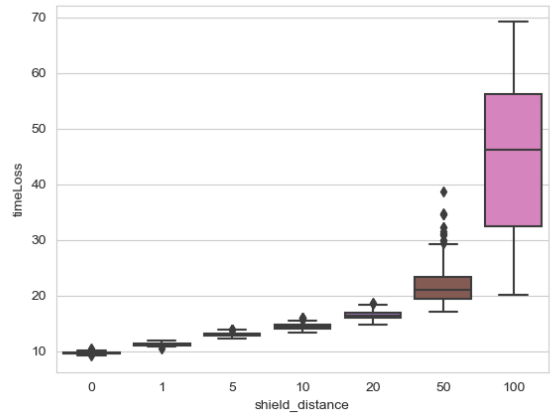


Figure 9: Time Loss over Shield Distance

collision detection. Our experiments have shown that changes in speed limits influence safety and performance in unexpected ways. Exploring causality seems worthwhile in this regard since the precise mechanisms remain unclear. Evaluating agents under different traffic patterns or adjusting the density of traffic within the entire simulation could provide valuable insights. Lastly, expanding our research to include other types of intersections or even entire road networks holds promise for further understanding and optimizing traffic management strategies.

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